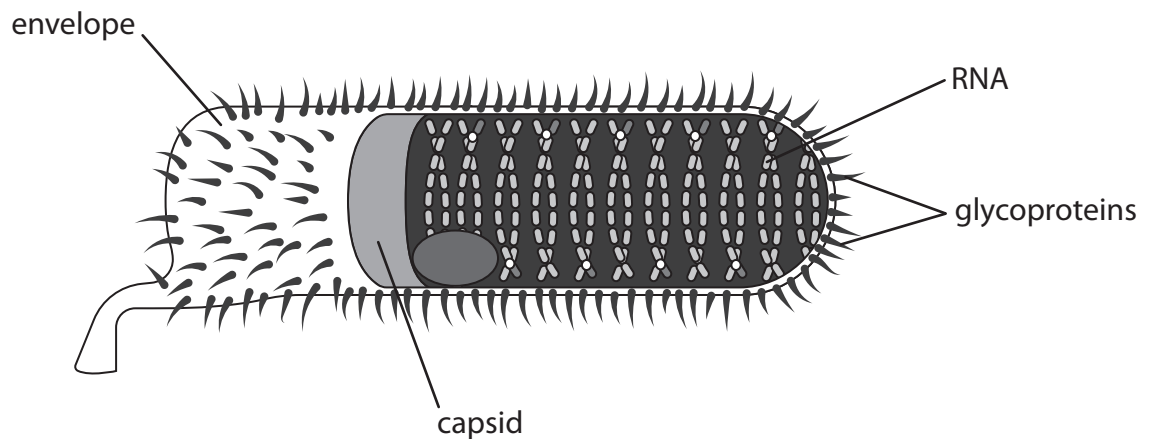


2 Rabies is a disease caused by a virus.

(a) The diagram shows the structure of the rabies virus.



(i) The rabies virus has an envelope.

Which of the following pairs of viruses have an envelope?

(1)

- ☐ **A** Ebola virus and human immunodeficiency virus (HIV)
- ☐ **B** human immunodeficiency virus (HIV) and tobacco mosaic virus (TMV)
- ☐ **C** lambda phage (λ phage) and Ebola virus
- ☐ **D** tobacco mosaic virus (TMV) and lambda phage (λ phage)

(ii) The structure of the rabies capsid is described as complex.

Which of the following has a complex capsid structure?

(1)

- ☐ **A** Ebola virus
- ☐ **B** human immunodeficiency virus (HIV)
- ☐ **C** lambda phage (λ phage)
- ☐ **D** tobacco mosaic virus (TMV)

(iii) Rabies virus is an RNA virus.

How many of the following viruses are RNA viruses?

- Ebola virus
- human immunodeficiency virus (HIV)
- lambda phage (λ phage)
- tobacco mosaic virus (TMV)

(1)

☐ **A** 1

☐ **B** 2

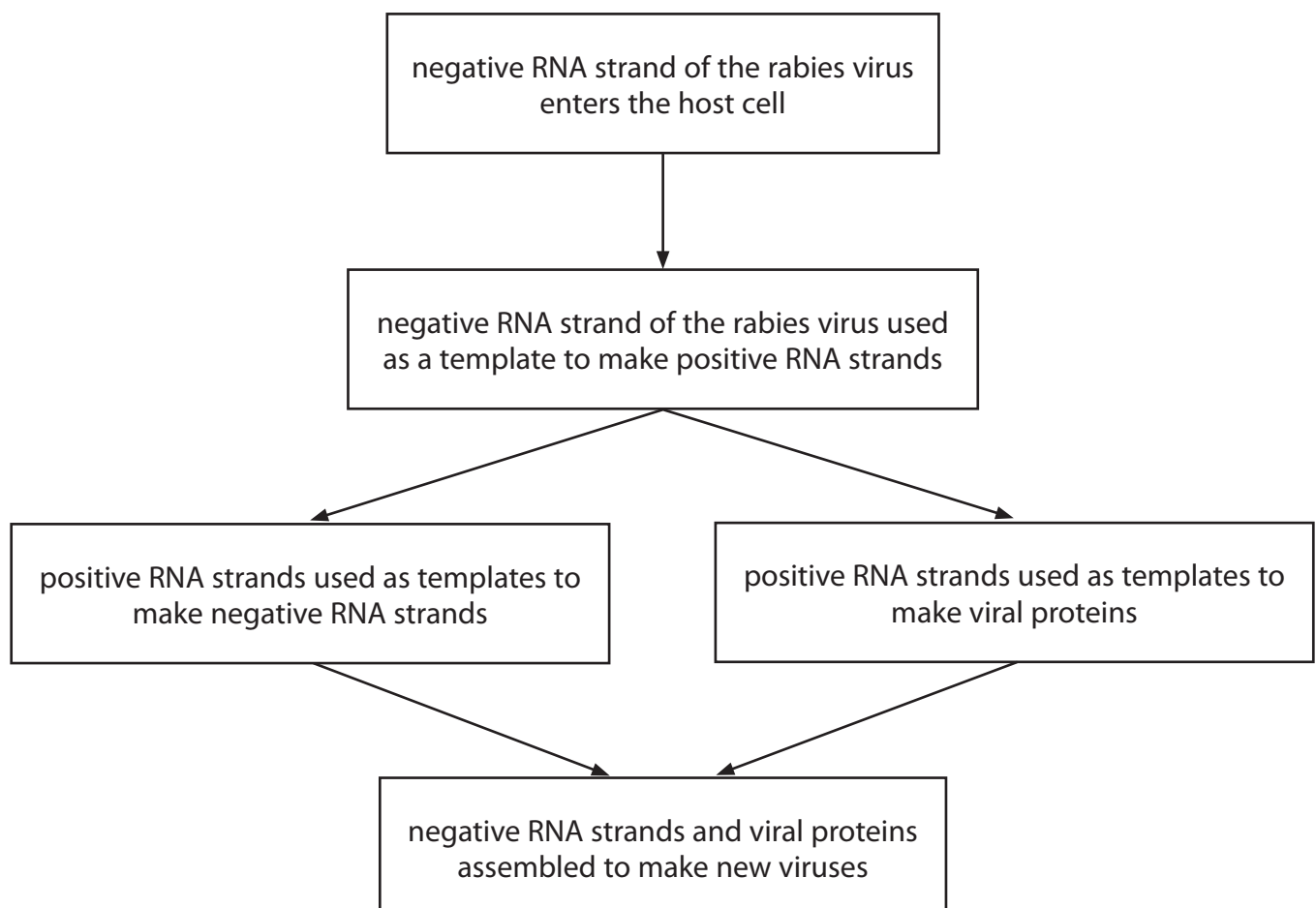
☐ **C** 3

☐ **D** 4

(b) The rabies virus replicates in a lytic cycle.

The RNA of the rabies virus is a negative RNA strand.

The diagram shows how the negative RNA strand of the rabies virus is used to make positive RNA and proteins.



- (i) The diagram shows part of the base sequence in the negative RNA strand.
- Complete the diagram to show the corresponding base sequence in the positive RNA strand.

(1)

Negative RNA strand	A	C	C	A	A	G	G	C	G
Positive RNA strand									

- (ii) Explain why a positive RNA strand has to be made.

(2)

(c) Lemurs are found in Madagascar. It is thought that they might carry rabies viruses.

The photograph shows a lemur.



(Source: Caroline Wilcox)

A person was bitten by a lemur.

This person did not receive any treatment for rabies until 18 days after being bitten.

Explain why doctors were worried that this person had left it too long for the treatment to be successful.

(4)

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(Total for Question 2 = 10 marks)